THIAMAZOLE

Agranulocytosis

Preferred outside 1st trimester

CHEMICAL NAME 1-methyl-1,3-dihydro-2H-imidazole-2-thione (methimazole)
CLASS Thionamide antithyroid drug
TRADE NAMES Tapazole · Thyrozol · Favistan · Metizol

THERAPEUTIC USES / INDICATIONS

- Hyperthyroidism (Graves', toxic multinodular goitre, toxic adenoma).
- Preparation for surgery/radioiodine.
- Thyroid storm (with PTU preferred in some protocols).

ADMINISTRATION

- **PO** in 3 divided doses about 8-hourly (US label); once-daily maintenance is possible owing to long intrathyroidal action.
- IV form available in some countries.

DOSAGES

- **Initial (US label):** 15 mg/day mild, 30–40 mg moderately severe, 60 mg severe, in 3 divided doses.
- **Maintenance:** 5–15 mg daily.
- **Children:** 0.4 mg/kg/day in 3 doses initially; maintenance about half.
- (Carbimazole is a prodrug of thiamazole; doses are not interchangeable mg-for-mg.)

PHARMACOKINETICS

- **Onset:** 1–2 weeks; euthyroid in 4–8 weeks.
- Concentrated in thyroid; metabolised in liver.
- Excreted in urine. **t½:** 4–6 h (effect 24 h).

PHARMACODYNAMICS

- Inhibits **thyroid peroxidase** → ↓ synthesis of T4/T3.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Rash, urticaria, pruritus.
- Arthralgia.
- Nausea, taste change.

ADVERSE EFFECTS (SERIOUS)

- Agranulocytosis.
- Cholestatic hepatitis.
- Vasculitis (ANCA-positive); rare.
- Teratogenicity (1st trimester).

CONTRAINDICATIONS & CAUTIONS

- Previous agranulocytosis or severe reaction.
- First trimester of pregnancy.
- Severe hepatic impairment.

KEY INTERACTIONS

- **Warfarin:** altered effect as thyroid status changes.
- **Digoxin, β-blockers, theophylline:** adjust doses as euthyroid.

NURSING MANAGEMENT

- Counsel on sore throat/fever; urgent FBC.
- TFTs every 4–6 weeks.
- LFTs if symptomatic.
- Pregnancy planning.

PATIENT EDUCATION

- Stop and seek urgent blood test for sore throat or fever.
- Report rash, yellow skin.
- Plan pregnancy with your doctor.

KEY POINTS

- Active thionamide (carbimazole metabolite).
- Agranulocytosis warning.
- Once daily.
- Switch to PTU in 1st trimester.

SOURCES CHECKED **[633]** Methimazole, oral (US label) **[634]** Methimazole, oral (US label, Bionpharma Inc)
Full references p. 432–457